

building management systems, energy modeling, day-light simulation, weather stations (which monitor the external conditions and calibrate the performance of buildings), online metering systems for energy and water, solar air-conditioning, task lighting, occulux sensors, daylight sensors, humidity sensors and variable-frequency drives.

All these electronics and IT components dovetail with the overall structure of the green building in realising the desired benefits. Solar PV panels are integrated aesthetically into the architecture of the building. These generate electricity, allow the entry of natural light, as well as insulate the building.

The building management system is a computer-based system that monitors and controls the building's mechanical and electrical equipment such as ventilation, lighting, power systems, fire systems and security systems.

Latest in green building deployments

Some of the features of green buildings include the use of a high-performance envelope, energy-efficient air-conditioning and lighting systems, local materials and 100 per cent onsite treatment of waste water.

CII-Sohrabji Godrej Green Business Centre, India's first LEED Platinum rated green building, became functional in 2004. It incorporates green features like the right orientation, wind towers, solar PV, intelligent building management system, CO2 sensors and Occulux.

This year, the Infosys Campus located in Hyderabad and Bayer's eco-commercial building in Noida have incorporated unique green features such as chilled beams for air-conditioning, solar PVs and enhanced lighting controls. Further, green buildings by Suzlon, One Earth, Turbo



Building-integrated photovoltaics are literally part of the structure of a building

Decline in Initial Incremental Cost Over the Years

Building	Year of green building award	Built-in area (sq.metres)	Rating achieved	Increase in cost	Payback Period
CII-Godrej GBC, Hyderabad	2003	1858	Platinum	18 per cent	7 years
ITC Green Centre, Gurgaon	2004	15,793	Platinum	15 per cent	6 years
Wipro, Gurgaon	2005	16,258	Platinum	8 per cent	5 years
Technopolis, Kolkata	2006	66,890	Gold	6 per cent	2 years
Spectral Services Consultants Office, Noida	2007	1393	Platinum	8 per cent	4 years
Hyderabad Institute of Technology And Management, Hyderabad	2007	7246	Silver	2 per cent	3 years
Kalpataru Square, Mumbai	2008	30,379	Platinum	2 per cent	2 years
Suzlon Energy, Pune	2010	76,180	Platinum	< 2 per cent	2 years
Kohinoor Hospital, Mumbai	2010	21,368	Platinum	< 2 per cent	2 years